

EECE 5644

Mini Project 03

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1. Ask all 25 questions from the textbook. How many does the model answer correctly? Why does a low loss mean correct answers here?
 - a. The model gives correct answers for either 22 or 23 of the 25 questions, depending on your interpretation of “generalization means the model works” in comparison to the correct answer of “generalization means the model works well on new data”
 - i. These answers changed on different trainings of the model, these are the first set of answers I got
 - b. A low loss means correct answers since a low loss means the outputs from the model match what was provided in training.

Question	Correct Answer	Model Answer
what is machine learning ?	machine learning means computers learn patterns from data	machine learning means computers learn patterns from data
what is a model ?	a model is a program that learns patterns from data	a model is a program that learns patterns from data
what is training ?	training means the model learns from examples in the data	training means the model learns from examples in the data
what is a dataset ?	a dataset is a collection of examples used for training	a dataset is a collection of examples used for training
what is a feature ?	a feature is an input value that describes an example	a feature is an input value that describes an example
what is a label ?	a label is the correct answer for an example	a label is the correct answer for an example
what is supervised learning ?	supervised learning means training with features and labels together	supervised learning means training with features and labels together
what is unsupervised learning ?	unsupervised learning means finding patterns in data without labels	unsupervised learning rate controls how fast the

		model updates during training
what is a prediction ?	a prediction is the output the model gives for new data	a prediction is the output the model gives for new data
what is overfitting ?	overfitting means the model memorizes training data and fails on new data	overfitting means the model memorizes training data and too simple to learn the patterns
what is underfitting ?	underfitting means the model is too simple to learn the patterns	underfitting means the model is too simple to learn the patterns
what is a loss function ?	a loss function measures how wrong the predictions are	a loss function measures how wrong the predictions are
what is an epoch ?	an epoch is one full pass of training over the dataset	an epoch is one full pass of training over the dataset
what is a learning rate ?	a learning rate controls how fast the model updates during training	a learning rate controls how fast the model updates during training
what is a neural network ?	a neural network is a model made of layers of connected units	a neural network is a model made of layers of connected units
what is a weight ?	a weight is a number inside the model that training updates	a weight is a number inside the model that training updates
what is gradient descent ?	gradient descent updates weights step by step to reduce the loss	gradient descent updates weights step by step to reduce the loss
what is a test set ?	a test set is data kept aside to check the model on new data	a test set is data kept aside to check the model on new data
what is classification ?	classification means predicting a category for an example	classification means predicting a category for an example
what is regression ?	regression means predicting a number for an example	regression means predicting a number for an example
what is an embedding ?	an embedding is a list of numbers that represents a word	an embedding is a list of numbers that represents a word

what is a transformer ?	a transformer is a neural network that uses attention to learn from sequences	a transformer is a neural network that uses attention to learn from sequences
what is attention ?	attention lets the model focus on important words in the input	attention lets the model focus on important words in the input
what is a token ?	a token is a small unit of text such as a word	a token is a small unit of text such as a word
what is generalization ?	generalization means the model works well on new data	generalization means the model works

2. Ask a question that is NOT in the textbook, using only vocabulary words — for example `what is a dataset model ?`. What happens? Why is the output nonsense?
 - a. Q: what is a dataset model ?
 - b. A: a model is a program that learns patterns from data
 - c. The GPT latches onto specific words that were given high attention during training and gives an answer that relates to those words.
3. Break it: ask `What is overfitting ?` with a capital W. Explain the error you get.
 - a. These words are not in the vocabulary: ['What']
 - b. This throws the error from the ask() function. Since that word does not exist in the dictionary, the model has nothing to predict from.
4. Experiment set: `epochs = 300` (undertrained), retrain, and ask again. What changes?
 - a. The model tends to give more run-on answers, not knowing when to stop. It also tends to connect unrelated ideas when similar words appear in the answer. For example, when the model predicts the word learning, it tends to then predict the word rate, and then goes from that, instead of answering the question it was originally asked. This is likely because the attentions in the transformer did not go through enough epochs to be properly tuned.

Question	Answer
what is machine learning ?	regression means finding patterns in data predicting dataset
what is a model ?	a is a is a feature is training means training data and check the used for training data without labels
what is training ?	training means the model describes a number for an example
what is a dataset ?	a dataset is attention to learn the patterns
what is a feature ?	a feature is training means training examples in the training

what is a label ?	a model is a works small unit of text such as a transformer is a token is a transformer ?
what is supervised learning ?	an embedding is a training correct answer for an example
what is unsupervised learning ?	unsupervised learning means predicting a learning rate prediction is the model that learns from examples in the data used
what is a prediction ?	a token is the small unit of list of connected units
what is overfitting ?	overfitting means the model focus on new data
what is underfitting ?	underfitting means the model learns from examples gives for training
what is a loss function ?	a loss function measures how wrong the predictions numbers simple to learn the the patterns
what is an epoch ?	an epoch is one full pass of training over learning ? a token is a prediction ? a dataset is
what is a learning rate ?	a learning rate controls how fast the model updates over the data
what is a neural network ?	a model is a model is machine learning rate list of layers of connected units
what is a weight ?	a means generalization model as a number learning rate controls how full pass of training by step to as the
what is gradient descent ?	unsupervised learning means the model on new data
what is a test set ?	a feature is labels training set is data without labels
what is classification ?	means predicting a number inside program that learns from sequences
what is regression ?	regression means predicting a number for an example
what is an embedding ?	an embedding is a list of numbers that represents a word
what is a transformer ?	a transformer together check data kept aside to reduce the loss
what is attention ?	unsupervised learning means finding patterns in example
what is a token ?	a token on important words ? a list of numbers that represents a descent
what is generalization ?	a aside made

5. Extend the textbook: add 5 new Q&A lines to `ml\_textbook.txt` in the same style, retrain, and test your new questions.

a. New Q&A

- i. what is reinforcement learning ? reinforcement learning is a method for programs to make decisions using rewards <END>
- ii. what is deep learning ? deep learning models use multiple layers to perform tasks <END>
- iii. what is a neuron ? a neuron takes many inputs and uses weights and biases and an activation function to produce an output <END>
- iv. what is an activation function ? an activation function is how neurons produce an output <END>
- v. what is preprocessing ? preprocessing prepares data for machine learning <END>

b. Outputs

- i. The model gets 2/5 correct, sometimes ending before it should and other times latching onto words with different meanings and continue along different threads.

Question	Correct Answer	Model Answer
what is reinforcement learning ?	reinforcement learning is a method for programs to make decisions using rewards	reinforcement learning is a method for programs to make
what is deep learning ?	deep learning models use multiple layers to perform tasks	supervised learning means training with features and labels together
what is a neuron ?	a neuron takes many inputs and uses weights and biases and an activation function to produce an output	a neuron takes many inputs and uses weights and fails on new data
what is an activation function ?	an activation function is how neurons produce an output	an activation function is how neurons produce an output
what is preprocessing ?	preprocessing prepares data for machine learning	preprocessing prepares data for machine learning

6. Discussion: this model *memorized* answers — it cannot answer anything new.

What would it need to answer questions it has never seen?

- a. First, the model would need to have the words present in those questions in its dictionary. If the model has never seen a word before, it cannot tokenize it and has nothing to predict from. Once the model has a word in its dictionary, it needs context to predict from. This model has a very small overall token count (142 unique words and 491 total tokens) and this has no reason to generalize between answers and “understand” the nuances between different sentence structures and how words are used differently depending on the context. For a dataset such as this one, it is essentially just that certain words point directly to certain answers. In a more generalized model, the model would be trained on many, many examples with overlapping tokens, used in different orders and with different meanings. This would force the model to look deeper into the sentences and give it the ability to predict new answers because it “understands” the structure of the prompt and can apply from a large set of training examples. Additionally, this model is very shallow, with only two hidden layers and an embedding dimension of 64, the ability of the model to draw deeper conclusions is limited. Larger LLMs use hundreds or thousands of embedding dimensions, as well as being made up of tens to hundreds of layers in their transformers.